

Table 6: **Multi-view generation.** We evaluate additional variants of the baselines using reference camera trajectories and single-view input images of the RealEstate10K test set. We compute reconstruction metrics based on the subsequent frames for the low-resolution and upsampled high-resolution generations.

Method	Low-resolution			High-resolution		
	PSNR ($\uparrow$)	SSIM ($\uparrow$)	LPIPS ($\downarrow$)	PSNR ($\uparrow$)	SSIM ($\uparrow$)	LPIPS ($\downarrow$)
MotionCtrl (latents)	14.67	0.323	0.331	13.23	0.467	0.562
CameraCtrl (patches)	14.42	0.304	0.366	13.04	0.450	0.577
MotionCtrl (frozen)	14.59	0.308	0.340	13.11	0.455	0.573
Ours	17.23	0.534	0.211	14.90	0.499	0.499

Table 7: **Quality metrics evaluation with additional baselines.** We evaluate additional variants of the baselines using text prompts from the RealEstate10K and MSR-VTT test sets respectively.

Method	RealEstate10K			MSR-VTT		
	FID ($\downarrow$)	FVD ($\downarrow$)	CLIPSIM ($\uparrow$)	FID ($\downarrow$)	FVD ($\downarrow$)	CLIPSIM ($\uparrow$)
MotionCtrl (latents)	1.83	77.39	0.2788	10.21	187.42	0.2636
CameraCtrl (patches)	2.57	71.04	0.2703	9.84	184.22	0.2612
MotionCtrl (frozen)	3.53	142.15	0.2772	8.19	165.48	0.2679
Ours	1.40	42.43	0.2807	7.80	165.18	0.2689

Table 8: **Camera pose evaluation for random rotation trajectories.** We evaluate all models using trajectories with randomly selected x-, y-, or z-axis and randomly sampled angular extent of the trajectory. We compute translation and rotation errors based on estimated camera poses from generations using ParticleSfM (Zhao et al., 2022).

Method	RealEstate10K		MSR-VTT	
	TransError ($\downarrow$)	RotError ($\downarrow$)	TransError ($\downarrow$)	RotError ($\downarrow$)
MotionCtrl	0.396	0.087	0.417	0.120
CameraCtrl	0.381	0.092	0.433	0.138
Ours	0.202	0.037	0.265	0.044

Table 9: **Camera pose evaluation for random trajectories with translations and rotations.** We evaluate all models using 11 trajectories for 1000 prompts with trajectories that involve significant directional changes including both rotations and translations. We compute translation and rotation errors based on estimated camera poses from generations using ParticleSfM (Zhao et al., 2022).

Method	RealEstate10K		MSR-VTT	
	TransError ($\downarrow$)	RotError ($\downarrow$)	TransError ($\downarrow$)	RotError ($\downarrow$)
MotionCtrl	0.451	0.095	0.456	0.146
CameraCtrl	0.369	0.088	0.479	0.135
Ours	0.236	0.041	0.258	0.050

Table 10: **Camera pose evaluation with vanilla video DiT backbone.** We incorporate our approach into a pre-trained vanilla DiT model in the latent space of CogVideoX (Yang et al., 2024d). We compute translation and rotation errors based on estimated camera poses from generations using ParticleSfM (Zhao et al., 2022).

Method	RealEstate10K		MSR-VTT	
	TransError ($\downarrow$)	RotError ($\downarrow$)	TransError ($\downarrow$)	RotError ($\downarrow$)
MotionCtrl	0.501	0.145	0.602	0.152
CameraCtrl	0.513	0.138	0.559	0.195
Ours	0.421	0.056	0.486	0.047

Table 11: **Quality metrics evaluation with vanilla video DiT backbone.** We incorporate our approach into a pre-trained vanilla DiT model in the latent space of CogVideoX (Yang et al., 2024d). We evaluate additional variants of the baselines using text prompts from the RealEstate10K and MSR-VTT test sets respectively.

Method	RealEstate10K			MSR-VTT		
	FID ($\downarrow$)	FVD ($\downarrow$)	CLIPSIM ($\uparrow$)	FID ($\downarrow$)	FVD ($\downarrow$)	CLIPSIM ($\uparrow$)
MotionCtrl	1.37	44.62	0.2752	9.68	157.90	0.2684
CameraCtrl	2.13	53.72	0.2748	8.32	152.88	0.2723
Ours	1.21	38.57	0.2834	6.88	137.62	0.2790